C4.5 Read transactions

This section defines the state changes associated with the Read transaction group that are issued on the AR channel. The Read transactions are:

- *ReadNoSnoop*
- *ReadOnce* on page C4-192
- *ReadClean* on page C4-192
- *ReadNotSharedDirty* on page C4-193
- *ReadShared* on page C4-194
- *ReadUnique* on page C4-195.

C4.5.1 ReadNoSnoop

ReadNoSnoop is a read transaction that is used in a region of memory that is not shareable with other masters. The transaction response requirements are:

- the IsShared response must be deasserted
- the PassDirty response must be deasserted.

Table C4-1 shows the expected cache line state changes for the ReadNoSnoop transaction:

Table C4-1 Expected ReadNoSnoop cache line state changes

Transaction	Start state	RRESP[3:2]	Expected end state	Legal end state	
		IsShared/PassDirty		With Snoop Filter	No Snoop Filter
ReadNoSnoop	I	00	I, UC	I, UC, SC	I, UC, SC

Note

A ReadNoSnoop transaction does not indicate if the cache line is allocated or not after the transaction has completed.

Table C4-2 shows the other permitted cache line state changes for the ReadNoSnoop transaction:

Table C4-2 Other permitted ReadNoSnoop cache line state changes

Transaction	Start state	RRESP[3:2]	Expected end state	Legal end state	
		IsShared/PassDirty		With Snoop Filter	No Snoop Filter
ReadNoSnoop	UC	00	I, UC	I, UC, SC	I, UC, SC
	UD	00	UD	UD, SD	UD, SD
	SC	00	I, UC	I, UC, SC	I, UC, SC
	SD	00	UD	UD, SD	UD, SD

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C4.5.2 ReadOnce

ReadOnce is a read transaction that is used in a region of memory that is shareable with other masters. This transaction is used when a snapshot of the data is required. The location is not cached locally for future use.

The transaction response requirements are:

- the IsShared response indicates if the cache line is shared or unique
- the PassDirty response must be deasserted.

Table C4-3 shows the expected cache line state changes for the ReadOnce transaction:

Table C4-3 Expected ReadOnce cache line state changes

Transaction	Start state	RRESP[3:2] IsShared/PassDirty	Expected end state	Legal end state With Snoop Filter	Legal end state No Snoop Filter
ReadOnce	I	00	I	I	I
		10	I	I	I

Table C4-4 shows the other permitted cache line state changes for the ReadOnce transaction:

Table C4-4 Other permitted ReadOnce cache line state changes

Transaction	Start state	RRESP[3:2] IsShared/PassDirty	Expected end state	Legal end state With Snoop Filter	Legal end state No Snoop Filter
ReadOnce	UC	00	UC	UC, SC	I, UC, SC
	UD	00	UD	UD, SD	UD, SD
	SC	00	UC	UC, SC	I, UC, SC
		10	SC	SC	I, SC
	SD	00	UD	UD, SD	UD, SD
		10	SD	SD	SD

C4.5.3 ReadClean

ReadClean is a read transaction that is used in a region of memory that is shareable with other masters. A ReadClean transaction is guaranteed not to pass responsibility for updating main memory to the initiating master.

Typically, a ReadClean transaction is used by a master that wants to obtain a clean copy of a cache line, for example a master with a write-through cache.

The transaction response requirements are:

- the IsShared response indicates if the cache line is shared or unique
- the PassDirty response must be deasserted.

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Table C4-5 shows the expected cache line state changes for the ReadClean transaction:

Table C4-5 Expected ReadClean cache line state changes

Transaction	Start state	RRESP[3:2] IsShared/PassDirty	Expected end state	Legal end state With Snoop Filter	Legal end state No Snoop Filter
ReadClean	I	00	UC	UC, SC	I, UC, SC
		10	SC	SC	I, SC

Table C4-6 shows other permitted cache line state changes for the ReadClean transaction:

Table C4-6 Other permitted ReadClean cache line state changes

Transaction	Start state	RRESP[3:2] IsShared/PassDirty	Expected end state	Legal end state With Snoop Filter	Legal end state No Snoop Filter
ReadClean	UC	00	UC	UC, SC	I, UC, SC
	UD	00	UD	UD, SD	UD, SD
	SC	00	UC	UC, SC	I, UC, SC
		10	SC	SC	I, SC
	SD	00	UD	UD, SD	UD, SD
		10	SD	SD	SD

C4.5.4 ReadNotSharedDirty

ReadNotSharedDirty is a read transaction that is used in a region of memory that is shareable with other masters. A ReadNotSharedDirty transaction can complete with any combination of the IsShared and PassDirty responses with the exception of both IsShared and PassDirty asserted.

Typically, the transaction is used by a cached master that is carrying out a load operation and can accept the cache line in any state except the SharedDirty state.

The transaction response requirements are:

- the IsShared response indicates if the cache line is shared or unique
- the PassDirty response indicates if the cache line is clean or dirty
- if the IsShared response indicates that the cache line is shared, then the PassDirty response must indicate that the cache line is clean.

Table C4-7 shows the expected cache line state changes for the ReadNotSharedDirty transaction:

Table C4-7 Expected ReadNotSharedDirty cache line state changes

Transaction	Start state	RRESP[3:2] IsShared/PassDirty	Expected end state	Legal end state With Snoop Filter	Legal end state No Snoop Filter
ReadNotSharedDirty	I	00	UC	UC, SC	I, UC, SC
		01	UD	UD, SD	UD, SD
		10	SC	SC	I, SC

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Table C4-8 shows other permitted cache line state changes for the ReadNotSharedDirty transaction:

Table C4-8 Other permitted ReadNotSharedDirty cache line state changes

Transaction	Start state	RRESP[3:2] IsShared/PassDirty	Expected end state	Legal end state With Snoop Filter	Legal end state No Snoop Filter
ReadNotSharedDirty	UC	00	UC	UC, SC	I, UC, SC
	UD	00	UD	UD, SD	UD, SD
	SC	00	UC	UC, SC	I, UC, SC
		01	UD	UD, SD	UD, SD
		10	SC	SC	I, SC
	SD	00	UD	UD, SD	UD, SD
		10	SD	SD	SD

Note

If a cache line starts in the SharedClean state, and the transaction response has PassDirty asserted, the cache line must move to a Dirty state.

C4.5.5 ReadShared

ReadShared is a read transaction that is used in a region of memory that is shareable with other masters. A ReadShared transaction can complete with any combination of the IsShared and PassDirty responses.

Typically, the ReadShared transaction is used by a cached master that is carrying out a load operation and can accept the cache line in any state.

The transaction response requirements are:

- the IsShared response indicates if the cache line is shared or unique
- the PassDirty response indicates if the cache line is clean or dirty.

Table C4-9 shows the expected cache line state changes for the ReadShared transaction:

Table C4-9 Expected ReadShared cache line state changes

Transaction	Start state	RRESP[3:2] IsShared/PassDirty	Expected end state	Legal end state With Snoop Filter	Legal end state No Snoop Filter
ReadShared	I	00	UC	UC, SC	I, UC, SC
		01	UD	UD, SD	UD, SD
		10	SC	SC	I, SC
		11	SD	SD	SD

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Table C4-10 shows other permitted state changes for the ReadShared transaction:

Table C4-10 Other permitted ReadShared cache line state changes

Transaction	Start state	RRESP[3:2] IsShared/PassDirty	Expected end state	Legal end state With Snoop Filter	Legal end state No Snoop Filter
ReadShared	UC	00	UC	UC, SC	I, UC, SC
	UD	00	UD	UD, SD	UD, SD
	SC	00	UC	UC, SC	I, UC, SC
		01	UD	UD, SD	UD, SD
		10	SC	SC	I, SC
		11	SD	SD	SD
	SD	00	UD	UD, SD	UD, SD
		10	SD	SD	SD

Note

If a cache line starts in the SharedClean state, and the transaction response has PassDirty asserted, the cache line must move to a Dirty state.

C4.5.6 ReadUnique

A ReadUnique transaction is used in a region of memory that is shareable with other masters. The transaction gets a copy of the data and also ensures that the cache line can be held in a Unique state. This permits the master to carry out a store operation to the cache line.

Typically, a ReadUnique transaction is used when the initiating master is carrying out a partial cache line store and does not have a copy of the cache line.

The transaction response requirements are:

- the IsShared response must be deasserted to indicate that the cache line is unique
- the PassDirty response must indicate if the cache line is clean or dirty.

Note

The cache line state changes associated with the ReadUnique transaction that Table C4-11 and Table C4-12 on page C4-196 show, do not include the cache line state changes associated with any subsequent store operation by the master once the cache line is in a Unique state.

Table C4-11 shows the expected cache line state changes for the ReadUnique transaction:

Table C4-11 Expected ReadUnique cache line state changes

Transaction	Start state	RRESP[3:2] IsShared/PassDirty	Expected end state	Legal end state With Snoop Filter	Legal end state No Snoop Filter
ReadUnique	I	00	UC	UC, SC	I, UC, SC
		01	UD	UD, SD	UD, SD

Table C4-12 on page C4-196 shows other permitted cache line state changes for the ReadUnique transaction:

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Table C4-12 Other permitted ReadUnique cache line state changes

Transaction	Start state	RRESP[3:2] IsShared/PassDirty	Expected end state	Legal end state	
				With Snoop Filter	No Snoop Filter
ReadUnique	UC	00	UC	UC, SC	I, UC, SC
	UD	00	UD	UD, SD	UD, SD
	SC	00	UC	UC, SC	I, UC, SC
		01	UD	UD, SD	UD, SD
	SD	00	UD	UD, SD	UD, SD

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